

Diego Pilares Gallego

dpilares01@gmail.com • dpilares01.github.io • linkedin.com/in/diegopilares/

Colmenar Viejo, Madrid, Spain • +34 689 525 579

Education

Oxford Brookes University

Oxford, UK

MSc in Motorsport Engineering. Merit

September 2024 – November 2025

Thesis: Hydrogen-Fuelled LTC/CAI-IC Engine Control Strategy Optimisation

Relevant Modules: • Advanced Vehicle Dynamics • Laptime Simulation and Race Engineering
• Advanced Vehicle Aerodynamics • Composite Design and Impact Modelling

Universidad Francisco de Vitoria – Motor & Sport Institute

Alcorcón, Spain

Beng in Industrial Engineering (Automotive specialisation)

September 2019 – January 2025

Thesis: Mono-Nut Carbon Fibre Wheel Design for Formula Student

Relevant Modules: • Suspension • Transmission and Gearbox • Powertrain • Machines and Mechanisms Theory
• Thermodynamics • Design, Prototyping and Testing

Experience

Oxford Brookes Racing

Oxford, UK

Formula Student Team Member – Suspension & Structures

September 2024 – May 2025

- Designed brake rotors and hub components using SolidWorks and HyperMesh.
- Applied topology optimisation to reduce mass while maintaining performance and reliability.

NTT DATA

Madrid, Spain

Industrial Engineer Placement – Automotive Department

February 2024 – September 2024

- Developed corrective measures and new functionalities for BMW online platforms across multiple international markets.
- Produced technical documentation and training materials.
- Analysed and resolved functional and data-related issues.
- Designed layouts for Hyundai Spain internal applications.

UFV Racing

Alcorcón, Spain

Formula Student Team Member – Electronics, Systems & Testing

September 2019 – July 2021

- Designed wiring diagrams, PCB layouts, and supported AIM PDM integration and engine mapping.
- Installed powertrain, suspension, steering, and brake systems.
- Produced CAD, 3D-printed and TIG welded test and final components.

Skills & Interests

Simulation: AVL VSM, AVL Drive Race, MoTeC i2, GT-Suite, MSC Adams, MATLAB/Simulink, Multisim

CAD, FEA & CFD: SolidWorks, CATIA V5, Star CCM+, XFLR5, Femap, HyperMesh

Productivity: Microsoft Office, Google Workspace, Jira

Language: Spanish (Native), English (C1, Professional working proficiency), German (A1, Basic)

Interests: Motorsports, Go-karting, Retro electronics restoration, Hands-on maintenance

Projects

MSc Dissertation - Hydrogen-Fuelled LTC/CAI-IC Engine Control Strategy Optimisation

Oxford Brookes University
January 2025 – September 2025

- Developed 1D LTC engine simulations using GT-Suite.
- Applied ML-based control optimisation to maximise engine performance and reduce NOx emissions.

Next-Generation F1 Compliance and Setup Study

Oxford Brookes University
April 2025 – May 2025

- Validated 2026-spec F1 vehicle setups using AVL VSM and AVL Drive.
- Optimised suspension, tyre behaviour, and brake balance.

LMGT3 DrivAer Front Aero Device Design and CFD Validation

Oxford Brookes University
October 2024 – December 2024

- Designed front aerodynamic components in SolidWorks.
- Performed CFD validation using Star CCM+, achieving improved aerodynamic performance.

Suspension Dynamics and Ride Optimisation

Oxford Brookes University
October 2024 – December 2024

- Developed bicycle, twin-track, and 1 to 4 DoF vehicle models in MSC Adams.
- Analysed cornering response and vertical load transfer behaviour.

Additional Information

Portfolio: Complete project documentation and technical work available at dpilares01.github.io